Teardown findings, failed 33 kV cable joints, July to September 2026

Failure analysis by the field service team of Raychem RPG, November 2026.

> Six failed joints were cut open and checked for the four causes in the set order. Water entry was the cause in 3 joints, a preparation error in 2 and a connector defect in 1. No joint failed from overload. All 3 joints with water entry were installed in rain, and the work was not stopped when rain reached the open cable.

How the teardown was done

Each failed joint stayed in place until photos were taken. It was then cut open along its length, and the point where the fault path started was found. The four causes were checked in order: preparation error, water entry, connector defect and overload of the circuit. This failed joint teardown followed the failure analysis procedure of the field service team.

Findings by joint

Joint	Site	Months in service	Cause	What was seen
1	Substation feeder, site A	5	Water entry	Water marks under the outer sleeve. The sleeve end was not sealed.
2	Substation feeder, site A	5	Water entry	Same installer team and same week as joint 1. Moisture around the connector.
3	Industrial plant, site B	8	Preparation error	Carbon track that starts at a knife cut in the insulation.
4	Wind farm collector line, site C	3	Water entry	Installed in rain without a tent. Water inside the sleeve.
5	Industrial plant, site B	11	Connector defect	Gap in the crimp. Heat marks on the connector.
6	Utility feeder, site D	14	Preparation error	Insulation not degreased. Tracking along the surface.

What the job records showed

5 of the 6 joints had a digital job card. 3 of these 5 had no photo of the sealed sleeve end, and these 3 were the joints with water entry. The second engineer had signed off all 5 records. The sign-off looked at the test readings and not at the photos.

Actions

Action	Owner	Due

Put the rule to stop work when rain or dust reaches the open cable on the first page of the jointing procedure	Head of field service	30 November 2026
Make the photo of the sealed sleeve end a required field of the digital job card, and add it to the sign-off of the second engineer	Quality lead	15 December 2026
Hold a refresher session on cable preparation for the installer teams of joints 3 and 6	Training lead	31 December 2026
Check the crimp tool of the installer team of joint 5 against its calibration record	Quality lead	30 November 2026

Synthetic sample written for the RPG NeuralMesh demonstration. It is not a record of Raychem RPG.